

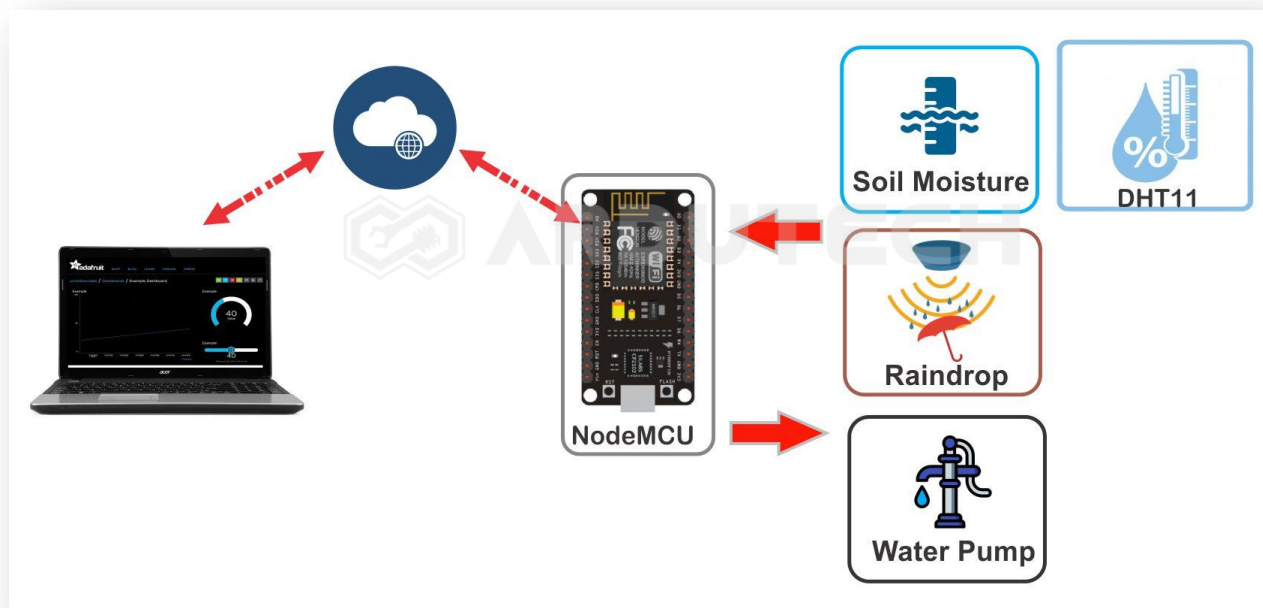
Smart Garden dengan Adafruit IO

Deskripsi

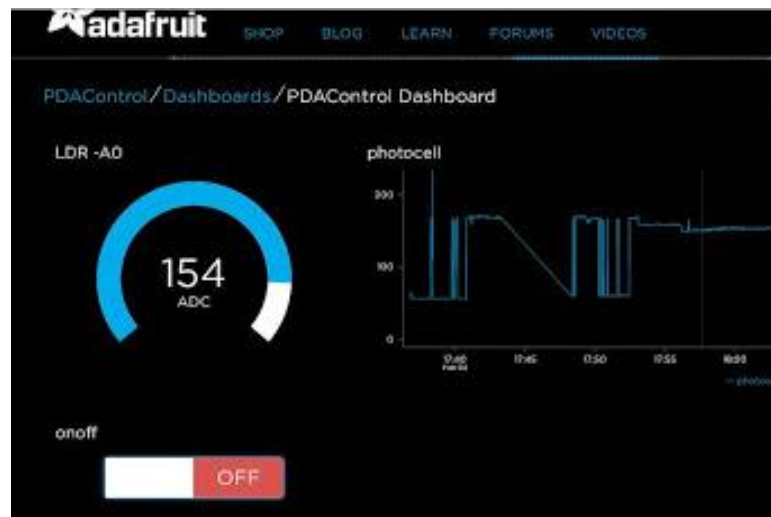
Membuat proyek IoT (*Internet of Things*) untuk monitoring dan controlling sistem perkebunan/tanaman seperti memantau kondisi kelembaban tanah, kondisi cuaca (hujan, kelembaban udara) dengan memakai platform IoT Adafruit IO.

Cara Kerja

Beberapa sensor seperti sensor suhu kelembaban udara DHT11/DHT22, sensor kelembaban tanah, sensor hujan membaca data lingkungan yang diukur kemudian data dikirim oleh NodeMCU melalui jaringan internet ke Adafruit secara kontinyu sehingga kita dapat mengetahui kondisi lingkungan tanah/perkebunan. Selain itu juga terhubung dengan sebuah pompa air untuk sistem penyiraman tanaman.

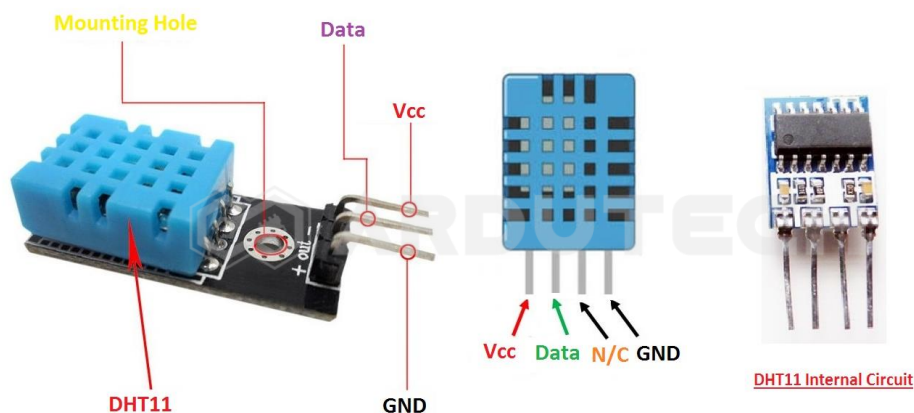


Adafruit IO adalah sistem yang disediakan oleh Adafruit (pembuat modul – modul sensor, perangkat elektronik, robot dll) untuk membuat sebuah proyek/aplikasi IoT (*Internet of Things*) . Adafruit IO ini boleh dipakai oleh siapa saja karena sifatnya free.



Sensor DHT11.

Sensor suhu & kelembaban DHT11 merupakan sensor paling banyak dipakai untuk aplikasi dasar monitoring suhu & kelembaban. Selain mudah didapatkan harganya juga relatif murah.



Spesifikasi sensor suhu kelembaban DHT11 :

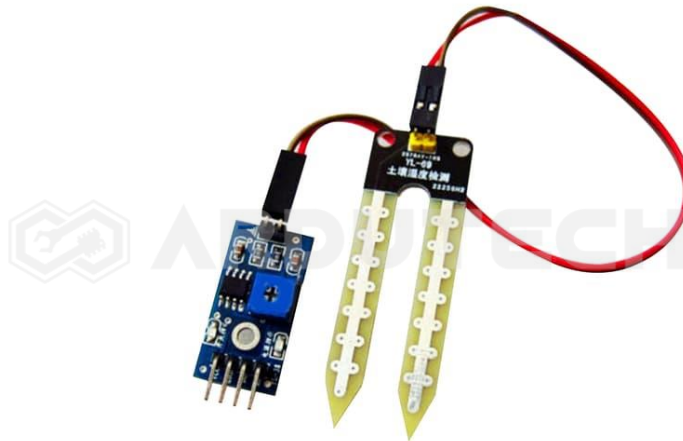
- Tegangan input : 3,5 – 5 VDC
- Sistem komunikasi : Serial (single – Wire Two way)
- Range suhu : 0°C – 50°C
- Range kelembaban : 20% - 90% RH
- Akurasi : $\pm 2^{\circ}\text{C}$ (temperature) $\pm 5\%$ RH (humidity)

Range pengukuran suhu memang hanya sampai 50 C, tetapi cukuplah untuk belajar sensor, toh suhu ruangan yang diukur biasanya tidak sampai lebih dari 50 C. Sensor suhu dan kelembaban DHT11 terdiri dari 4 kaki/pin, tetapi yang dipakai hanya 3 pin saja. Biasanya kalau kita membeli dalam bentuk modul jumlah pin-nya menjadi 3 :

- VCC(+) : tegangan input (5V)
- GND(-) : Ground
- DOUT : Data output serial

Sensor Kelembaban Tanah.

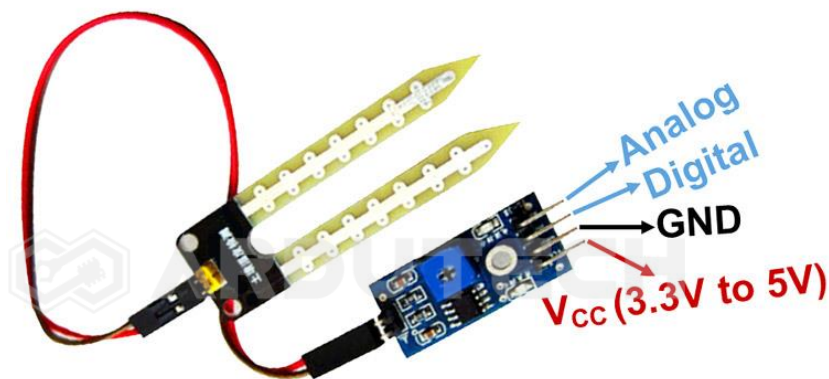
Sensor kelembaban tanah ini dipakai untuk mengukur kadar air/kelembaban di dalam tanah, jadi ada bagian sensor (elektroda) yang nantinya dimasukkan ke dalam tanah.



Pada sensor kelembaban tanah juga sudah dilengkapi dengan modul *signal conditioning* sehingga kita tinggal menghubungkan dengan NodeMCU tanpa perlu tambahan komponen lain.

Spesifikasi sensor kelembaban tanah :

- Chip comparator : LM393
- Output analog dan digital
- Tegangan input : 3.3 V – 5 VDC
- Terdapat trimpot untuk mengatur sensitifitas



Terdapat 4 pin/kaki pada modul sensor :

- VCC(+) : tegangan input (3.3 - 5V)
- GND(-) : Ground
- AO : Analog Output
- DO : Digital Output

Sensor Hujan (Raindrop).

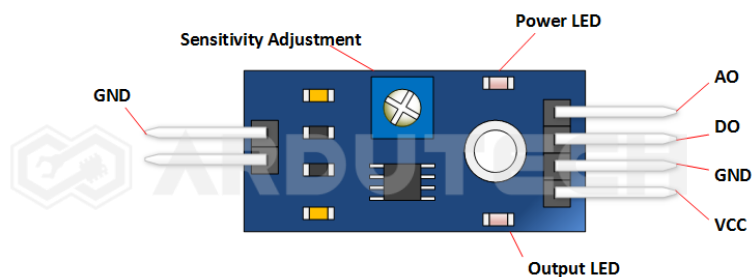
Sensor ini untuk mengukur rintik air hujan.



Ada 2 bagian : bagian detektor dan *signal conditioning*. Bagian detektornya berupa plat yang diletakkan menghadap ke-atas untuk menerima air hujan dengan ukuran 5 x 4 cm dengan lapisan nikel dan berbahan FR-04.

Spesifikasi sensor hujan :

- Tegangan input : 3.3 – 5 VDC
- IC Comparator LM393
- Terdapat 2 output yaitu Digital dan Analog
- Terdapat trimpot untuk mengatur sensitifitas



Pada modul signal conditioning terdapat 4 pin/kaki :

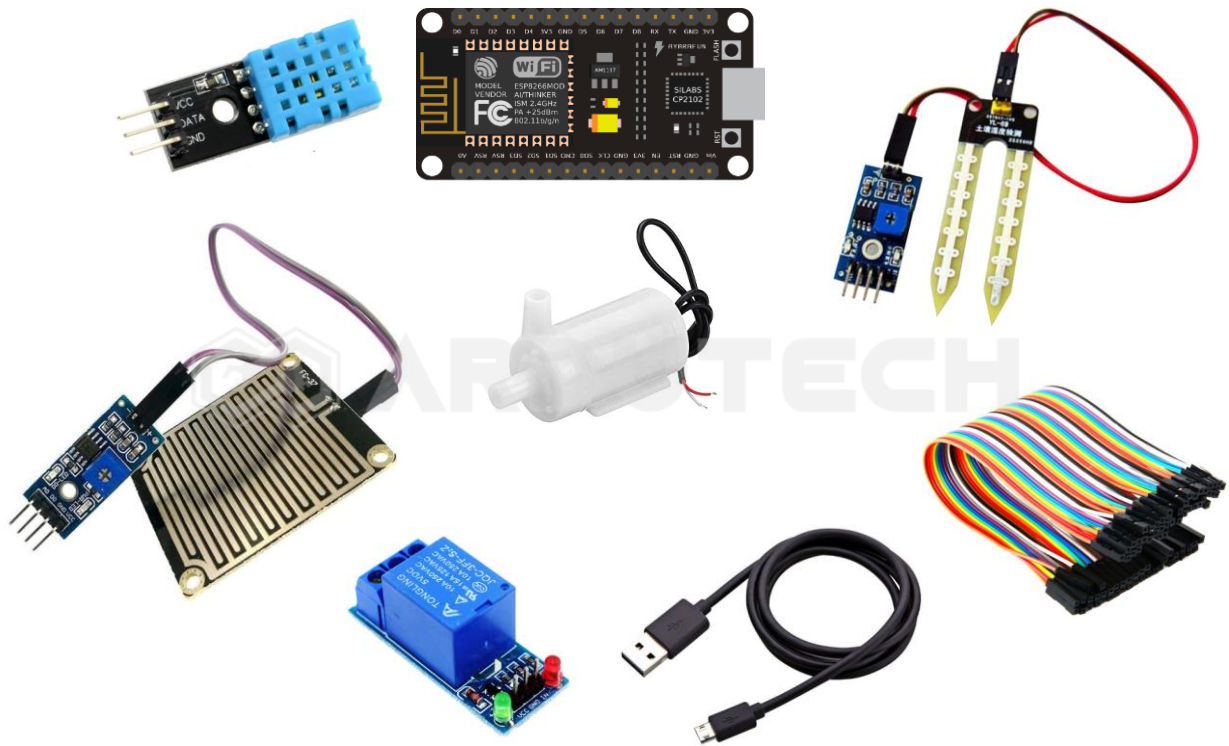
- VCC(+) : tegangan input (3.3 - 5V)
- GND(-) : Ground
- AO : Analog Output
- DO : Digital Output

Kebutuhan Bahan

- NodeMCU V3

www.ardutech.com @2020 (Terimakasih anda tidak meng-copy file ini untuk dijual kembali)

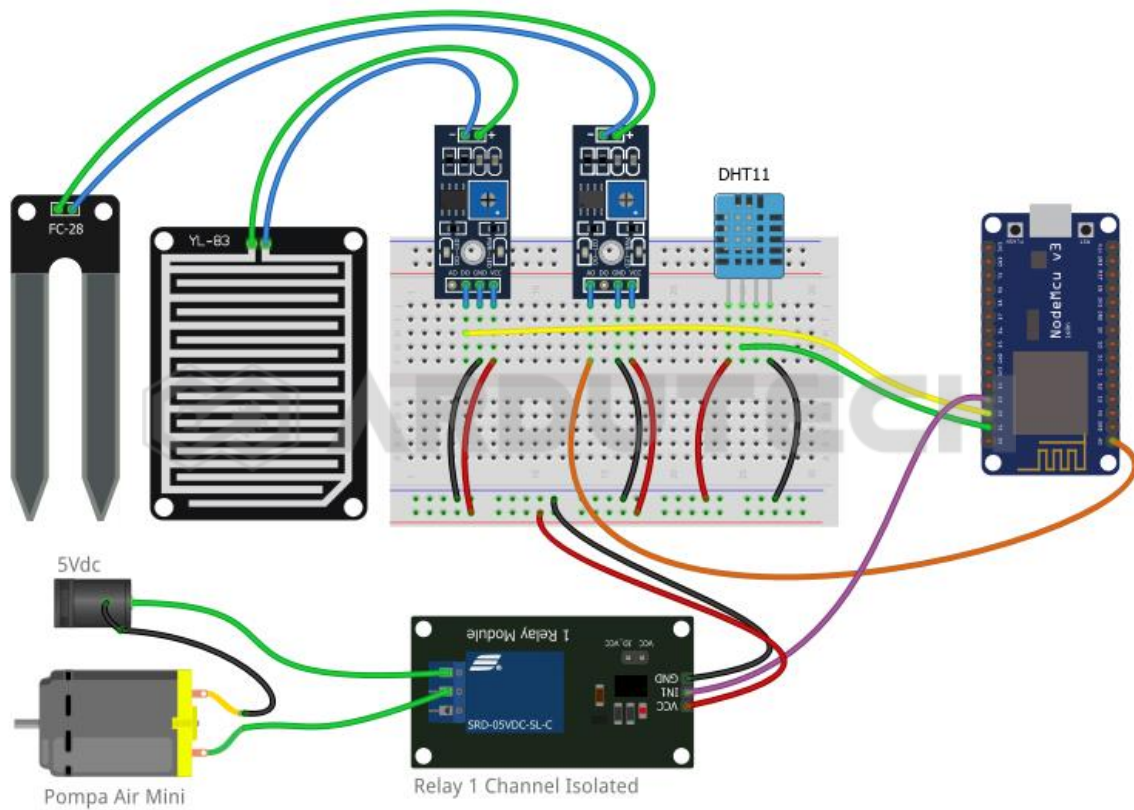
- Sensor suhu kelembaban DHT11
- Sensor kelembaban tanah (**soil moisture**)
- Sensor hujan (**raindrop**)
- Relay modul 1 Ch
- Pompa air (mini water pump)
- Kabel konektor
- Kabel micro USB



Kebutuhan Software

- Arduino IDE
- Adafruit IO

Rangkaian/Skematik



Koneksi NodeMCU dengan Sensor DHT11 :

NodeMCU	Sensor DHT11
D1	OUT
5V	VCC
GND	GND

Koneksi NodeMCU dengan Sensor Rain drops :

NodeMCU	Sensor Rain drops
D2	DO
5V	VCC
GND	GND

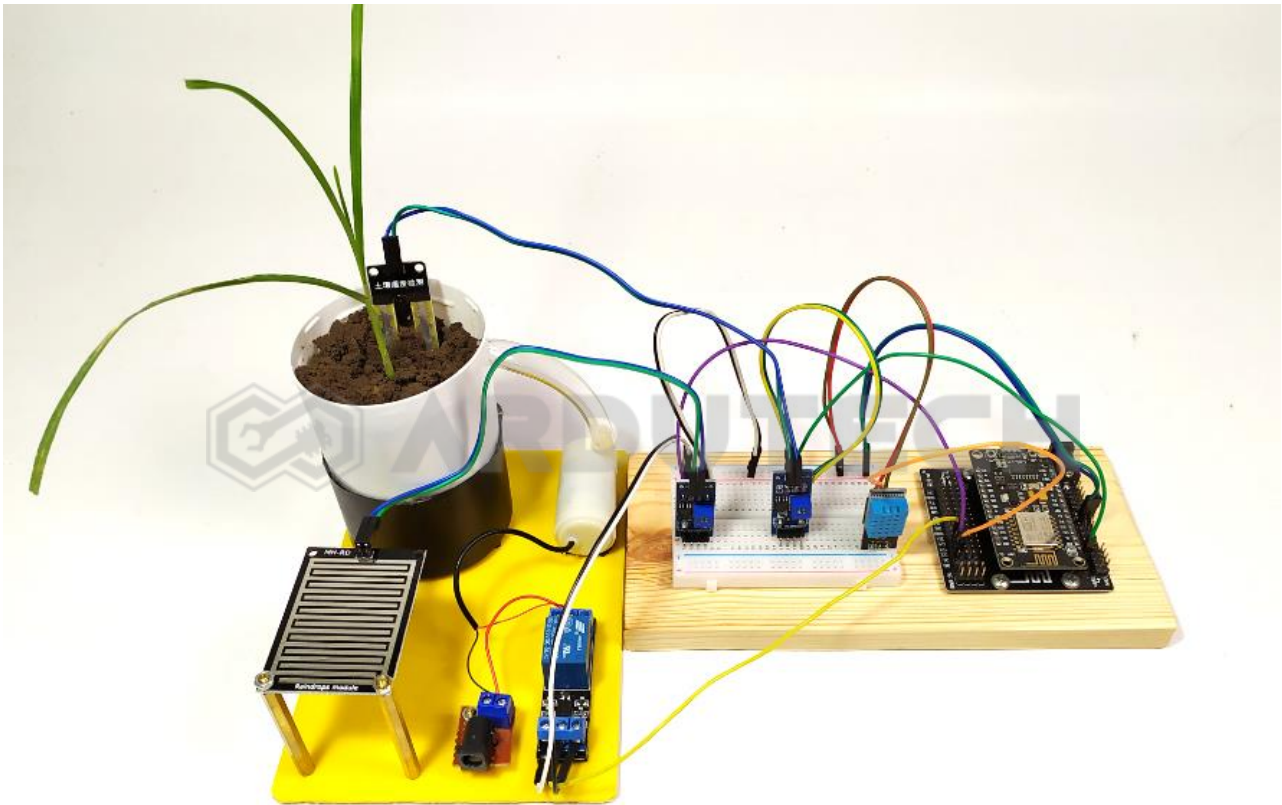
Koneksi NodeMCU dengan Sensor Soil Moisture :

NodeMCU	Sensor Soil Moisture
A0	AO
5V	VCC
GND	GND

Koneksi NodeMCU dengan Relay 1 Channel :

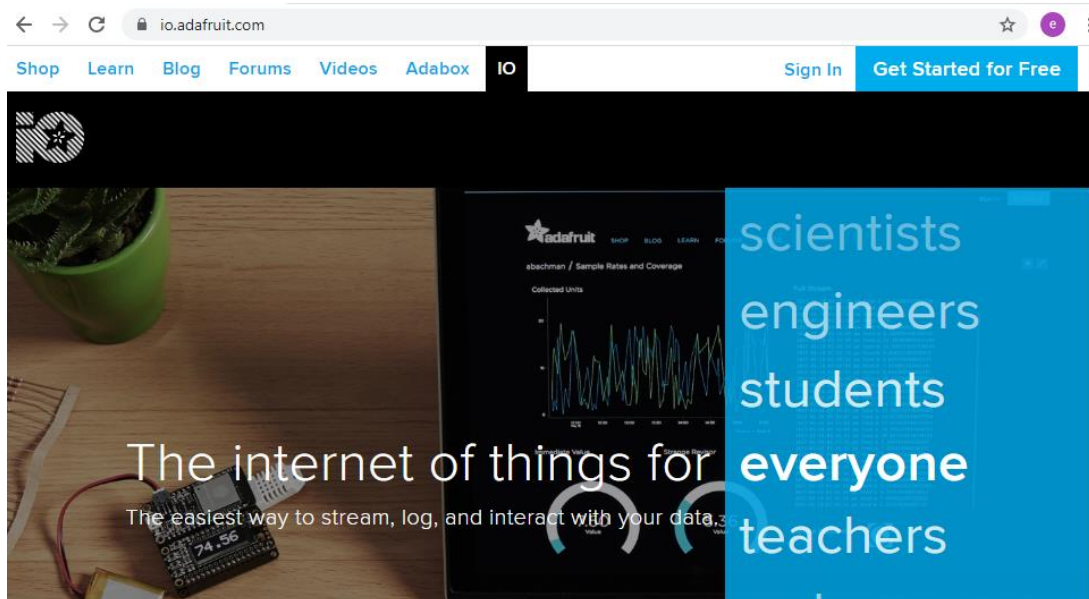
NodeMCU	Relay 1 Ch
D3	IN
5V	VCC
GND	GND

Relay yang digunakan **modul relay 1 channel isolated**.



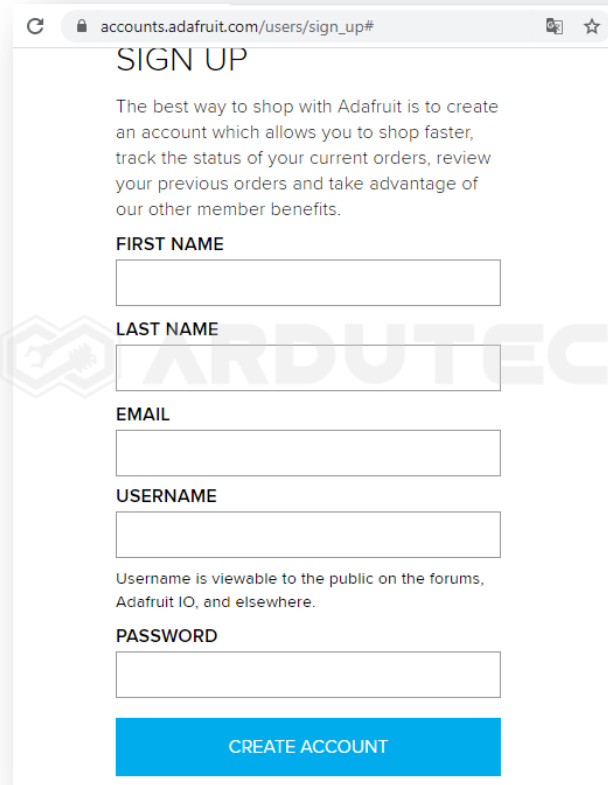
Membuat Antarmuka di Adafruit

Silakan masuk ke halaman <https://io.adafruit.com/>



Jika belum mempunyai akun silakan membuat akun terlebih dahulu. Siapkan sebuah akun email aktif.

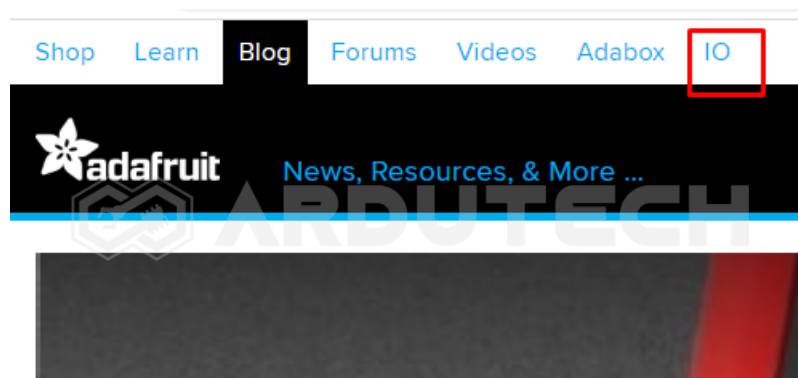
Klik pada **"Get Started for free"** di bagian pojok kanan atas.



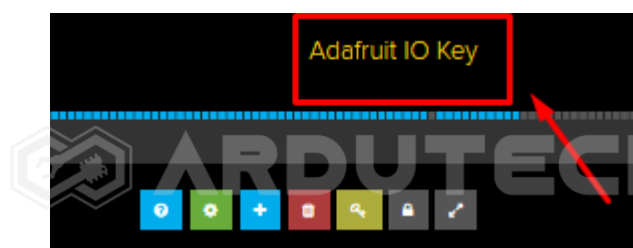
The screenshot shows the 'SIGN UP' page on accounts.adafruit.com. It includes a heading 'SIGN UP', a paragraph explaining the benefits of creating an account, and form fields for 'FIRST NAME', 'LAST NAME', 'EMAIL', 'USERNAME', and 'PASSWORD'. A blue 'CREATE ACCOUNT' button is at the bottom. A note states: 'Username is viewable to the public on the forums, Adafruit IO, and elsewhere.'

Isi data – data yang diperlukan kemudian klik **“CREATE ACCOUNT”** dan selanjutnya ikuti petunjuknya.

Masuk ke menu **IO**.



Selanjutnya kita perlu kode (token) untuk mengisi variabel username dan active key di program Arduino yang akan dibuat. Klik bagian **“Adafruit IO Key”** di pojok kanan atas.



YOUR ADAFRUIT IO KEY

Your Adafruit IO Key should be kept in a safe place and treated with the same care as your Adafruit username and password. People who have access to your Adafruit IO Key can view all of your data, create new feeds for your account, and manipulate your active feeds.

If you need to regenerate a new Adafruit IO Key, all of your existing programs and scripts will need to be manually changed to the new key.



Username

ArdutechJogja

Active Key

ruu_wUXJ670DJNEIIBQWERTY567MNB

REGENERATE KEY

[Hide Code Samples](#)

Arduino

```
#define IO_USERNAME "ArdutechJogja"
#define IO_KEY      "ruu_wUXJ670DJNEIIBQWERTY567MNB"
```

Perhatikan untuk **Username** dan **Active Key** ini penting, karena nanti dipakai ketika pemrograman dengan Arduino IDE. Catat/simpan kode tersebut. , untuk setingan awal Adafruit sampai disini dulu.

Program/Source Code

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h
- Adafruit_MQTT.h
- Adafruit_MQTT_Client.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/******
* Project Smart Garden dengan Adafruit
* Board : NodeMCU ESP8266 V3
* Input : Modul Soil Moisture
*        Modul Rain Drops
*        DHT11
* actuator: Relay --> pump
* Output : Blynk
* 99 Proyek IoT
* www.ardutech.com
```

```
*****/

#include <ESP8266WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
#include <Wire.h>
#include <DHT.h>

#define DHTPIN D1
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

#define SOILPIN A0
#define RAINPIN D2
#define PUMPPIN D3

byte humi, temp;
byte delaySend;
byte MODE=1;

//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
#define WLAN_SSID      "ArdutechWiFi" // Nama Hotspot/WiFi
#define WLAN_PASS      "12345678"     // Password
#define AIO_SERVER      "io.adafruit.com"
#define AIO_SERVERPORT  1883           // use 8883 for SSL
//---GANTI SESUAI DENGAN USER NAME ADAFRUIT ANDA
#define AIO_USERNAME    "ArdutechJogja"
//---GANTI SESUAI DENGAN IO KEY ADAFRUIT ANDA
#define AIO_KEY          "aio_mAXJ90DJnEIBVj0IDmG6BIMZqMAW"

/***** Global State (you don't need to change this!)
*****/
```

```

WiFiClient client;

Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT,
AIO_USERNAME, AIO_USERNAME, AIO_KEY);

/***** Feeds *****/
*****/

Adafruit_MQTT_Publish TEMP = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/temp DHT22");

Adafruit_MQTT_Publish HUMI = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/humi DHT22");

Adafruit_MQTT_Publish SOIL = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/soilmoisture");

Adafruit_MQTT_Publish RAIN = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/Raindrop");

Adafruit_MQTT_Subscribe MANUAL = Adafruit_MQTT_Subscribe(&mqtt,
AIO_USERNAME "/feeds/manual pump", MQTT_QOS_1);

Adafruit_MQTT_Subscribe PUMP = Adafruit_MQTT_Subscribe(&mqtt,
AIO_USERNAME "/feeds/onoffpump", MQTT_QOS_1);

void onoffMANUAL(char *data, uint16_t len) {
  String message = String(data);
  message.trim();
  if (message == "ON") {MODE=2;Serial.println("MODE: 2");}
  if (message == "OFF") {MODE=1;Serial.println("MODE: 1");}
}

void onoffPUMP(char *data, uint16_t len) {
  Serial.print("PUMP: ");
  Serial.println(data);
  String message = String(data);
  message.trim();
  if (message == "ON" && MODE==2) {digitalWrite(PUMPPIN, LOW);} //RELAY
  AKTIF LOW
  if (message == "OFF" && MODE==2) {digitalWrite(PUMPPIN, HIGH);}

```

```
}

void setup() {
  pinMode(PUMPPIN, OUTPUT);
  digitalWrite(PUMPPIN, LOW);
  pinMode(RAINPIN, INPUT_PULLUP);
  Serial.begin(9600);
  delay(10);
  Serial.println();
  Serial.println();
  Serial.println("SMART GARDEN");
  // Connect to WiFi access point.
  Serial.print("Connecting to ");
  Serial.println(SSID);
  WiFi.begin(SSID, PASS);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println();
  Serial.println("WiFi connected");
  Serial.println("IP address: "); Serial.println(WiFi.localIP());
  dht.begin();
  MANUAL.setCallback(onoffMANUAL);
  mqtt.subscribe(&MANUAL);
  PUMP.setCallback(onoffPUMP);
  mqtt.subscribe(&PUMP);
}

void loop() {
  MQTT_connect();
  humi = dht.readHumidity();
```

```
temp = dht.readTemperature();
if (isnan(humi) || isnan(temp)) {
    Serial.println("DHT22 tidak terbaca... !");
    return;
}

word soilMoisture=analogRead(SOILPIN);
soilMoisture=map(soilMoisture, 0, 1023, 100, 0);

if(MODE==1){
    if(soilMoisture<25 && digitalRead(RAINPIN)==1){
        digitalWrite(PUMPPIN, LOW);
    }
    else {
        digitalWrite(PUMPPIN, HIGH);
    }
}

Serial.print("Mode: ");
if(MODE==1) Serial.println("Otomatis");
else Serial.println("Manual");
Serial.print("Pump: ");
if(digitalRead(PUMPPIN)) Serial.println("OFF");
else Serial.println("ON");
//-----SEND TEMP
Serial.print("Sending temperature: ");
Serial.print(temp);
Serial.print(" -->");
if (! TEMP.publish(temp)) {
    Serial.println("Failed");
} else {
    Serial.println("OK!");
}
```



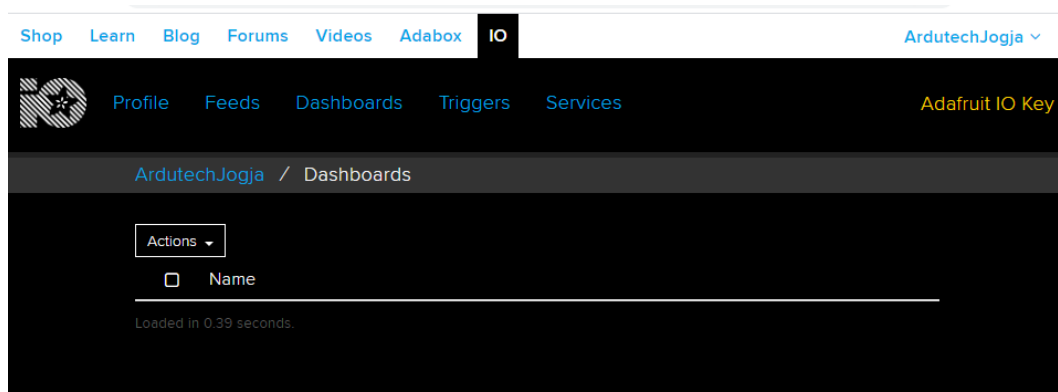
```
}  
  
//-----SEND HUMI  
Serial.print("Sending humidity: ");  
Serial.print(humi);  
Serial.print(" -->");  
if (! HUMI.publish(humi)) {  
    Serial.println("Failed");  
} else {  
    Serial.println("OK!");  
}  
  
//-----SEND SOIL MOISTURE  
Serial.print("Sending soil moisture: ");  
Serial.print(soilMoisture);  
Serial.print(" -->");  
if (! SOIL.publish(soilMoisture)) {  
    Serial.println("Failed");  
} else {  
    Serial.println("OK!");  
}  
  
//-----SEND RAINDROP STATUS  
Serial.print("Sending rain drop status: ");  
Serial.print(digitalRead(RAINPIN));  
Serial.print(" -->");  
if (! RAIN.publish(digitalRead(RAINPIN))) {  
    Serial.println("Failed");  
} else {  
    Serial.println("OK!");  
}  
  
mqtt.processPackets(10000);  
Serial.println();  
Serial.println();
```

```
}

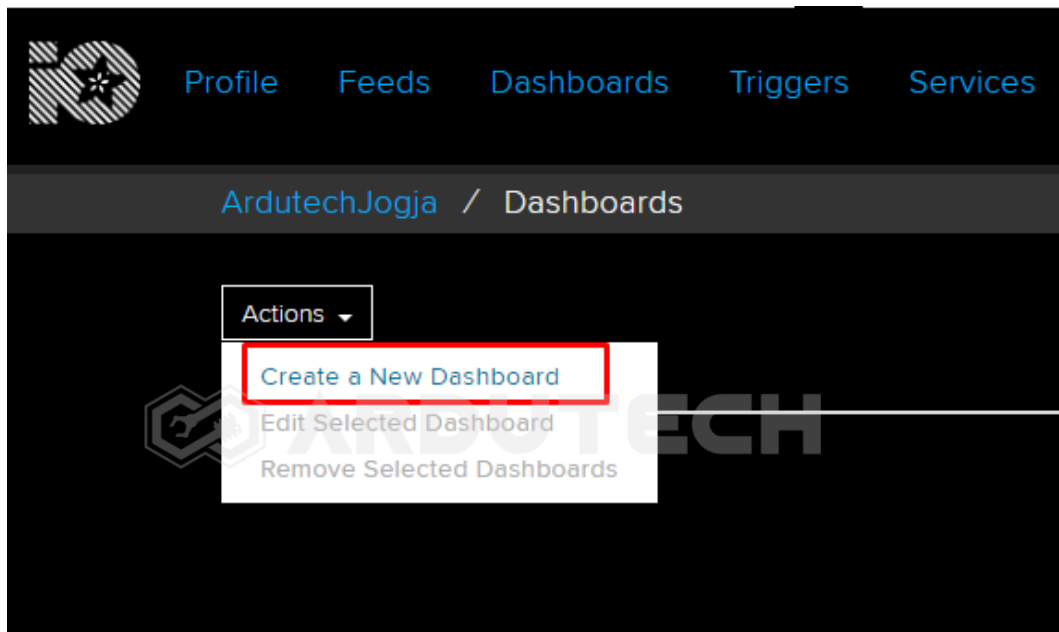
void MQTT_connect() {
    int8_t ret;
    if (mqtt.connected()) {
        return;
    }
    Serial.print("Connecting to MQTT... ");
    uint8_t retries = 3;
    while ((ret = mqtt.connect()) != 0) {
        Serial.println(mqtt.connectErrorString(ret));
        Serial.println("Retrying MQTT connection in 10 seconds...");
        mqtt.disconnect();
        delay(10000); // wait 10 seconds
        retries--;
        if (retries == 0) {
            while (1);
        }
    }
    Serial.println("MQTT Connected!");
}
```

Simpan (Save) kemudian Upload.

Selanjutnya masuk ke Adafruit IO, buat dashboard dengan nama "Smart Garden"



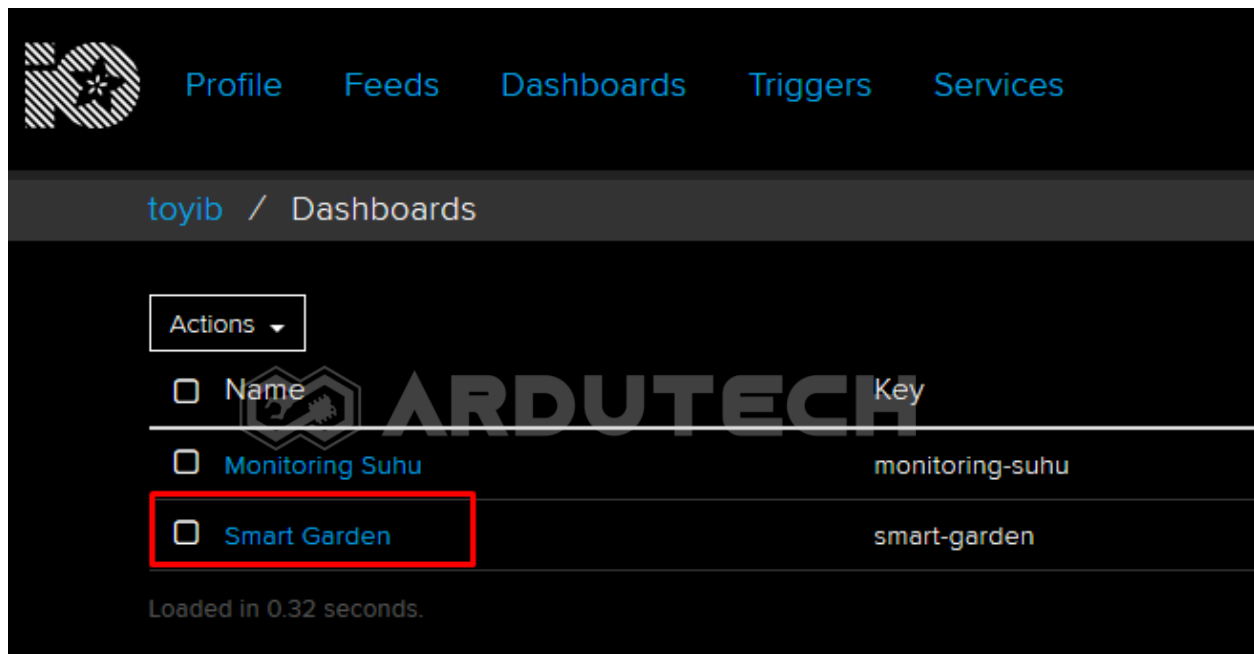
Klik pada tombol "Actions" kemudian pilih "Create a New Dashboard"



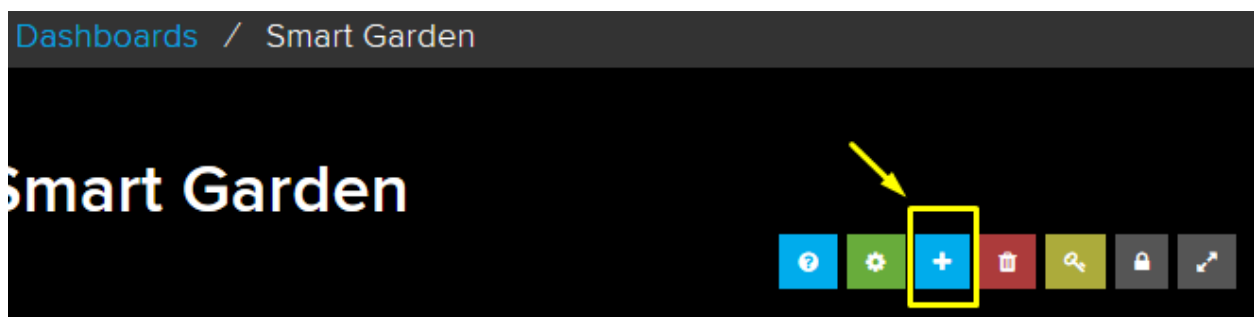
Selanjutnya akan muncul jendela untuk memberi nama, silakan beri nama pada dashboard yang akan dibuat, misalnya **“Smart Garden”** dan klik **“Create”**.

A screenshot of a modal window titled 'Create a new Dashboard'. It features a close button (X) in the top right corner. The form contains two input fields: 'Name' with the text 'Smart Garden' entered, and 'Description' which is currently empty. At the bottom right of the modal, there are two buttons: 'Cancel' and 'Create'.

Setelah klik tombol **“Create”** , akan muncul dashboard baru dengan nama **“Smart Garden”**.



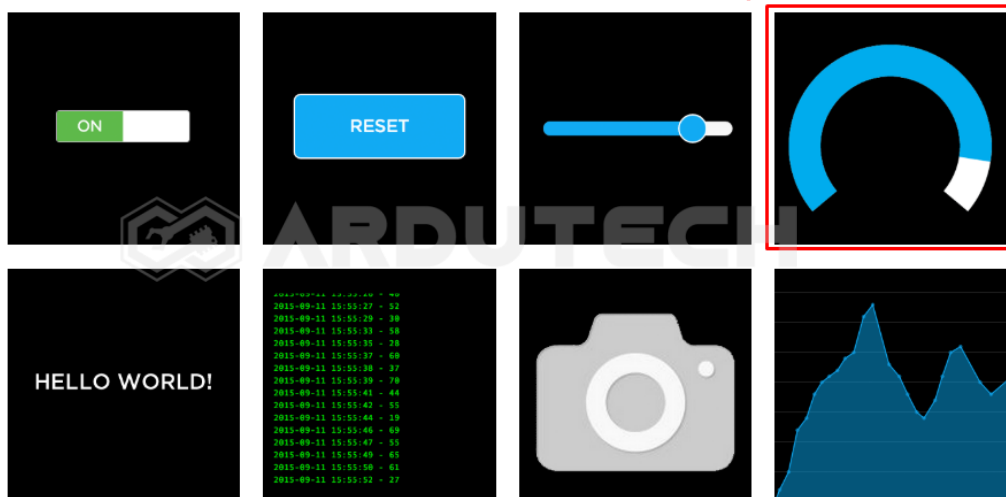
Klik “Smart Garden” kemudian pilih tombol “+” untuk membuat blok baru.



Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih ‘Chose feed’. Ini semacam “Topik” dalam komunikasi MQTT. Jadi supaya dikenali oleh ‘subscriber’ (penerima perintah) maka ‘publisher’ harus menentukan ‘topik’ nya. Silakan tulis **“temperature”** kemudian klik tombol **“Create”**.

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
--------------	------------	----------

Muncul feed temperature

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☐ LED	ON3	about 12 hours
☐ Light	0	about 13 hours
☐ photocell	561	about 12 hours
☒ temperature	33.55	2 minutes

Selanjutnya klik **"Next Step >"** sehingga muncul tampilan **Block settings** :

button to send it to your dashboard.

Block Title (optional)

Gauge Min Value

Gauge Max Value

Gauge Width

Gauge Label

Low warning value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Number of decimal places to display when value is a number. Defaults to 2.

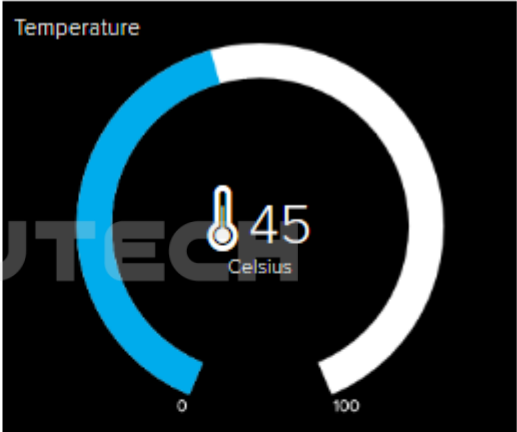
☒ Show Icon

When checked, show an icon with the value.

Icon

Show this icon next to the value.

Block Preview



Gauge A gauge is a read only block type that shows a fixed range of values.

Test Value

Isi pada kolom yang tersedia :

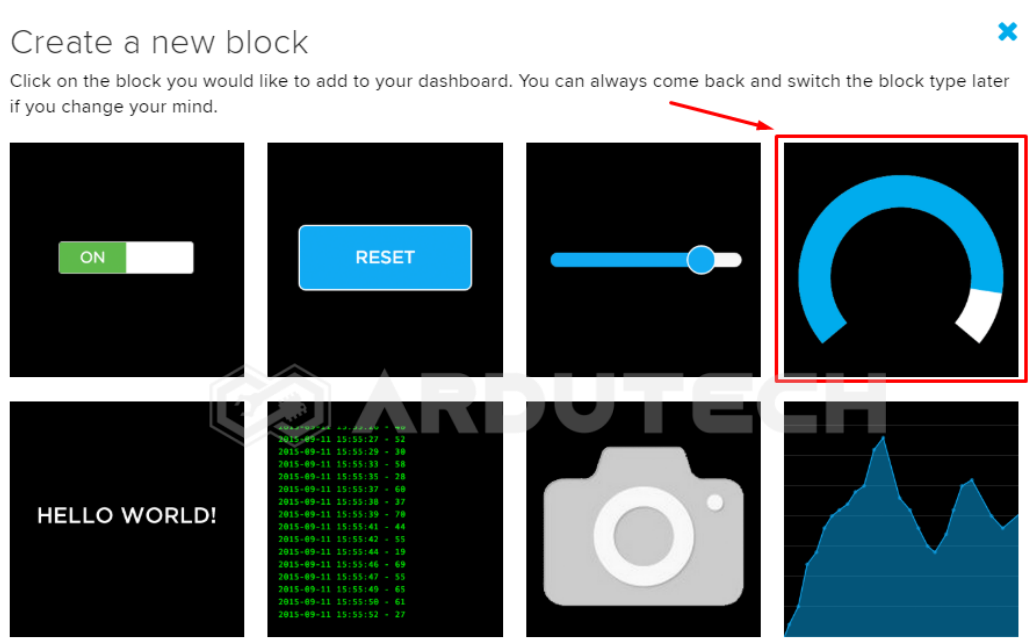
- Block Title : Temperature (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : Celcius
- Icon : w:thermometer

Klik tombol **“Create block”**.

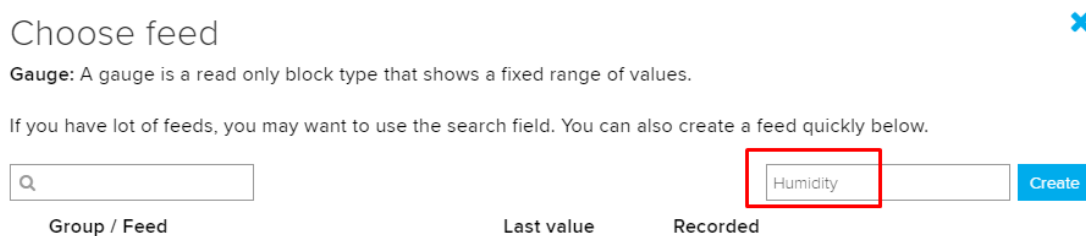


Klik tombol plus “+” lagi untuk menambahkan sebuah Gauge lagi ke dashboard.

Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).



Selanjutnya muncul tampilan untuk memilih ‘Chose feed’. Ini semacam “Topik” dalam komunikasi MQTT. Jadi supaya dikenali oleh ‘subscriber’ (penerima perintah) maka ‘publisher’ harus menentukan ‘topik’ nya. Silakan tulis “**Humidity**” kemudian klik tombol “**Create**”.



Muncul feed temperature.

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
☒ Humidity		less than a minute
☐ LED	ON3	about 14 hours
☐ temperature	33.55	10 minutes

< Previous step Next step >

Selanjutnya klik “**Next Step** > ” sehingga muncul tampilan **Block settings** :

Block Title (optional)

Humidity

Gauge Min Value

0

Gauge Max Value

100

Gauge Width

25px

Gauge Label

Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

2

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon

When checked, show an icon with the value.

Icon

w:humidity

Show this icon next to the value.

Block Preview

Humidity

Gauge A gauge is a read only block type that shows a fixed range of values.

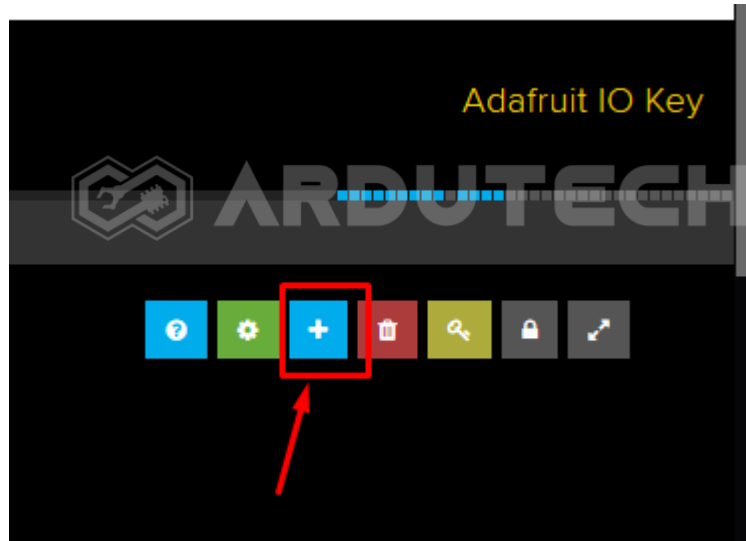
Test Value

45

Isi pada kolom yang tersedia :

- Block Title : Humidity (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Icon : w:humidity

Klik tombol “**Create block**”.



Pilih line chart, selanjutnya beri centang pada cek box soilmoisture. Kemudian pilih Next step

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.

The grid contains the following blocks:

- ON/OFF switch
- RESET button
- Slider control
- Progress bar
- HELLO WORLD! text
- Stream (line graph) - **Highlighted**
- Camera icon
- Arduino logo
- Blue circle
- Map
- Keypad
- Cloud icon
- Green circle
- Numeric keypad
- Text block: Hello world! This is a different type of text block that works for larger quantities of data.
- Streaming Data, Text Blocks, Welcome Dashboard, Shared Feeds list

Choose up to 5 feeds

Line Chart: The line chart is used to graph one or more feeds.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☐ Humidity	95	4 minutes
☐ manual pump	OFF	20 minutes
☐ onoffpump	OFF	20 minutes
☐ Raindrop	1	14 minutes
☒ soilmoisture	39	14 minutes
☐ Suhu LM35	34.80	3 months
☐ Temperature	29	3 minutes

1 of 5

Berikutnya masuk pada Block setting ganti keterangan dan parameter line chart:
Block Title : SOIL MOISTURE

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Show History

X-Axis Label

Y-Axis Label

Y-Axis Minimum

Leave blank to automatically detect.

Y-Axis Maximum

Leave blank to automatically detect.

Decimal Places

Number of decimal places to display, defaults to 4.

☐ Raw Data Only

When checked, do not show aggregate data ever. If the chart history includes more than 640 data points, only the 640 most recent will be shown.

☐ Stepped Line

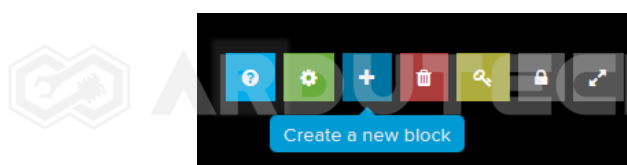
Use a stepped line graph. Useful for representing logic levels.

Block Preview

Line Chart The line chart is used to graph one or more feeds.

Jika sudah terisi klik Create block.

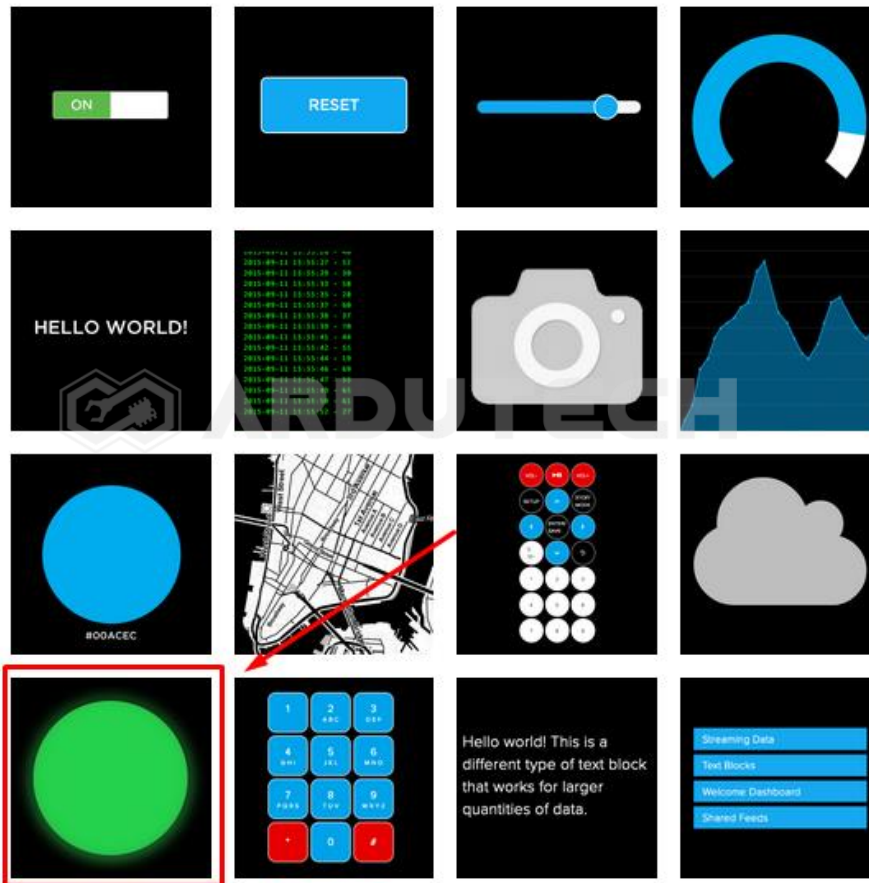
Selanjutnya sensor hujan, pilih button Create a new block (+)



Pilih Indicator, selanjutnya beri centang pada cek box Raindrop. Kemudian pilih Next step

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Choose feed

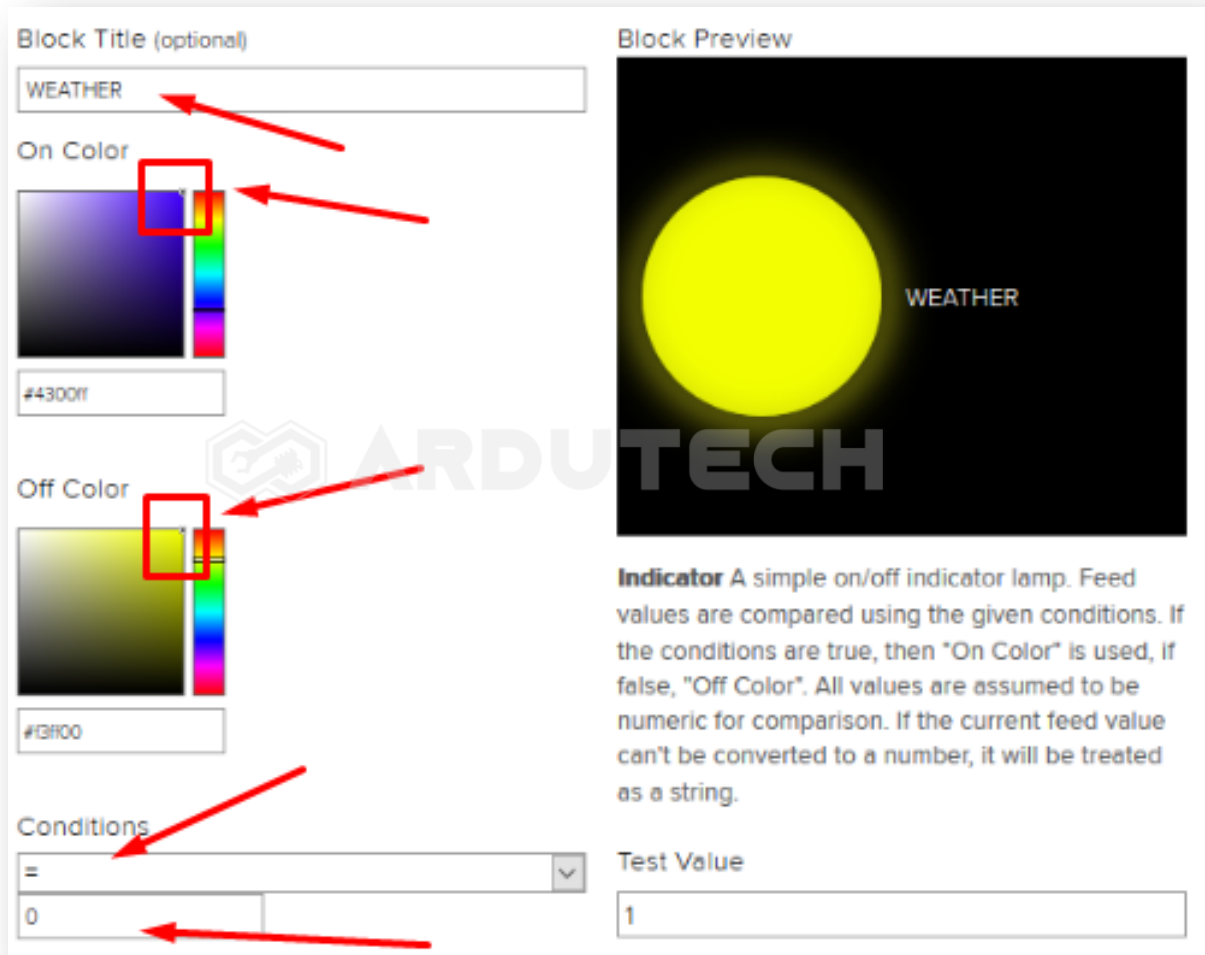
Indicator: A simple on/off indicator lamp. Feed values are compared using the given conditions. If the conditions are true, then "On Color" is used, if false, "Off Color". All values are assumed to be numeric for comparison. If the current feed value can't be converted to a number, it will be treated as a string.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☐ Humidity	95	10 minutes
☐ manual pump	OFF	25 minutes
☐ onoffpump	OFF	25 minutes
☒ Raindrop	1	20 minutes
☐ Soilmoisture	39	20 minutes
☐ Suhu LM35	34.80	3 months
☐ Temperature	29	9 minutes

Berikutnya masuk pada Block setting ganti keterangan dan parameter Indicator:

1. Block Title : WEATHER
2. On Color : Biru tua
3. Off Color : Kuning
4. Conditions : = 0



Block Title (optional)

WEATHER

On Color

#4300ff

Off Color

#ffff00

Conditions

=

0

Block Preview

WEATHER

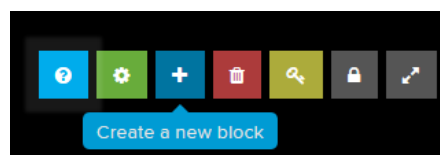
Indicator A simple on/off indicator lamp. Feed values are compared using the given conditions. If the conditions are true, then "On Color" is used, if false, "Off Color". All values are assumed to be numeric for comparison. If the current feed value can't be converted to a number, it will be treated as a string.

Test Value

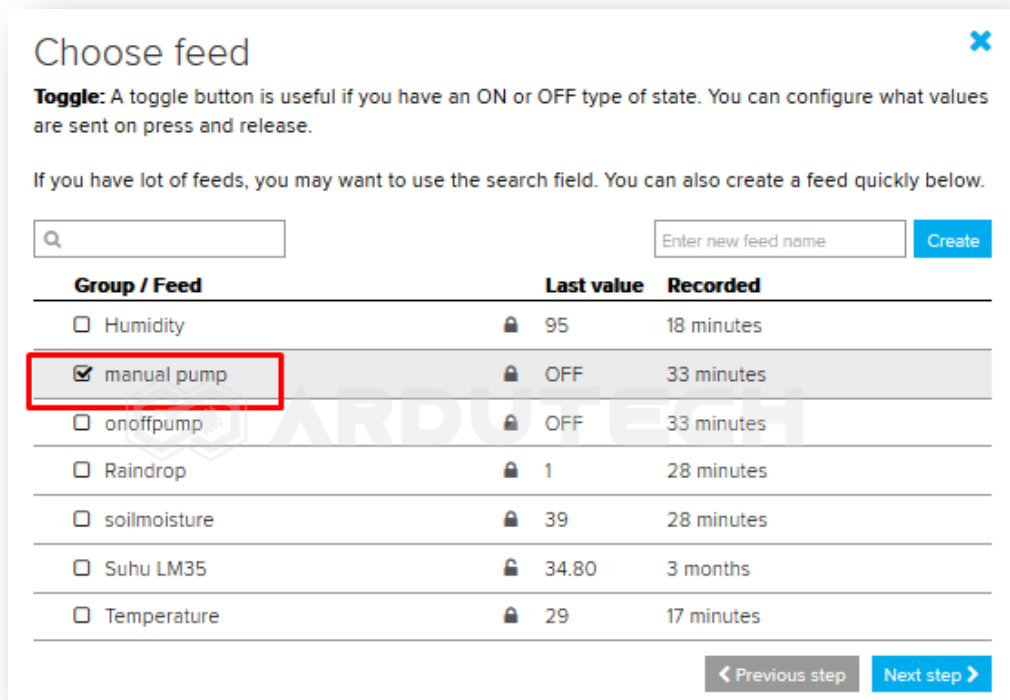
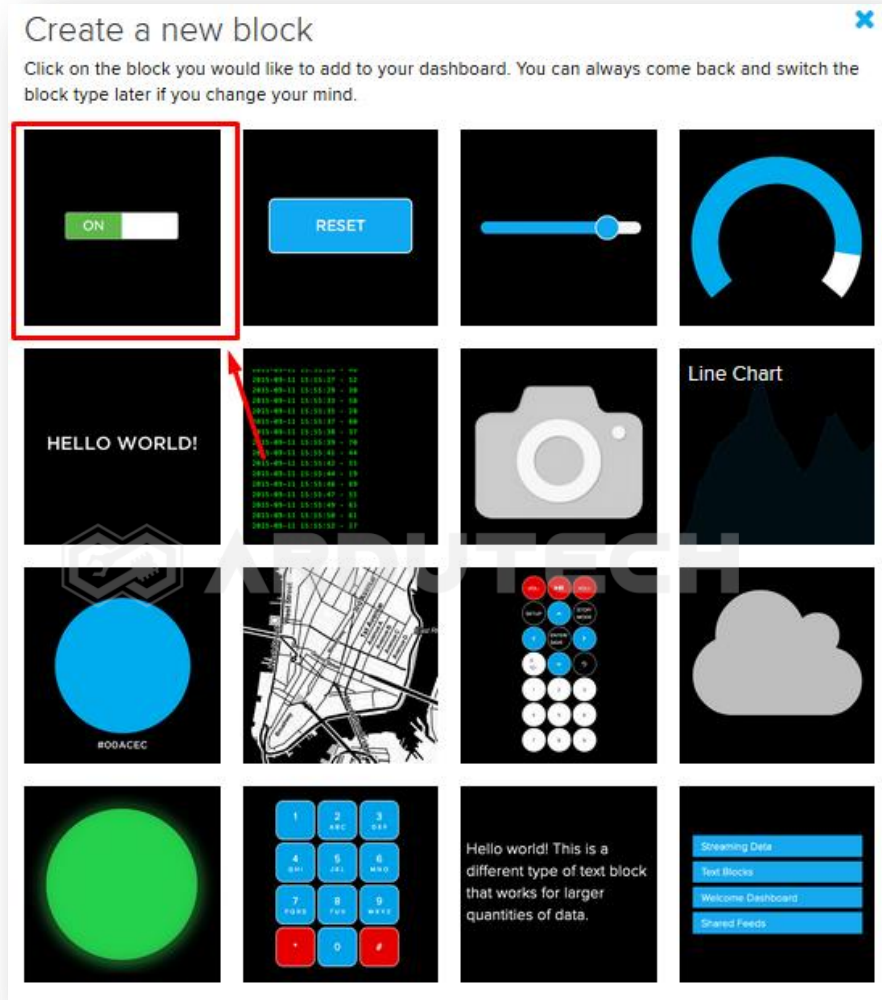
1

Jika sudah terisi klik Create block.

Selanjutnya tombol manual pump, pilih button Create a new block (+)



Pilih Toggle, selanjutnya beri centang pada cek box manual pump. Kemudian pilih Next step.



Berikutnya masuk pada Block setting ganti keterangan dan parameter Toggle:

Block Title : MANUAL PUMP

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)
MANUAL PUMP

Button On Text
ON

Button Off Text
OFF

Block Preview
MANUAL PUMP

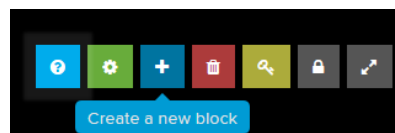
Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value
45

Published Value
0 bytes

Jika sudah terisi klik **Create block**.

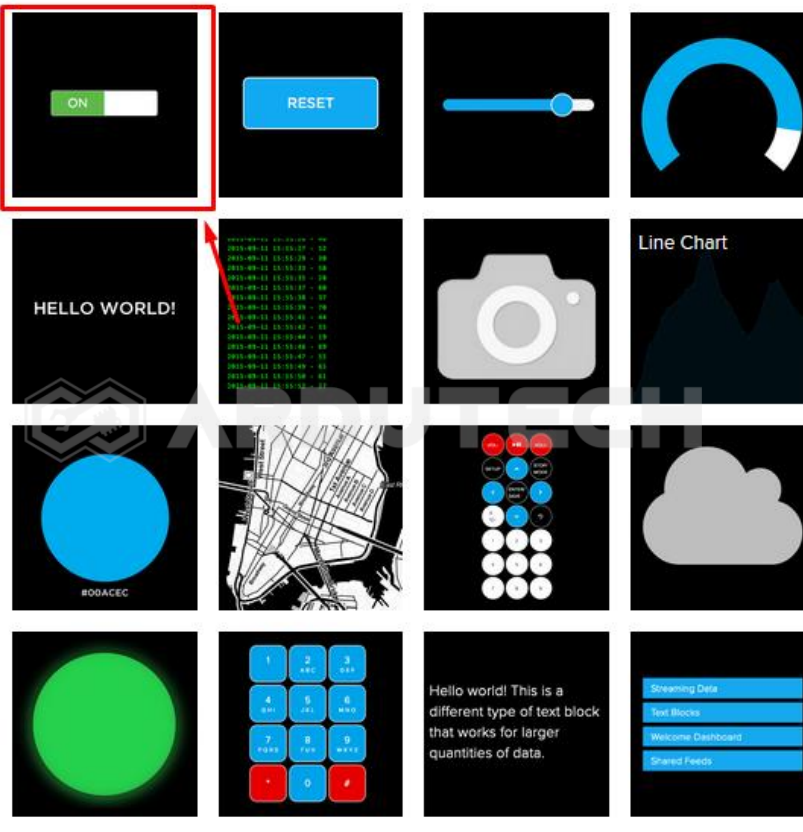
Selanjutnya tombol pump control, pilih button Create a new block (+)



Pilih Toggle, selanjutnya beri centang pada cek box onoffpump. Kemudian pilih Next step

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Choose feed

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☐ Humidity	95	22 minutes
☐ manual pump	OFF	38 minutes
☒ onoffpump	OFF	38 minutes
☐ Raindrop	1	32 minutes
☐ soilmoisture	39	32 minutes
☐ Suhu LM35	34.80	3 months
☐ Temperature	29	22 minutes

Berikutnya masuk pada Block setting ganti keterangan dan parameter Toggle:

Block Title : MANUAL PUMP

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)
PUMP CONTROL

Button On Text
ON

Button Off Text
OFF

Block Preview

PUMP CONTROL

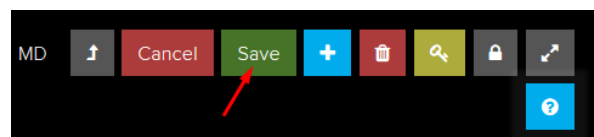
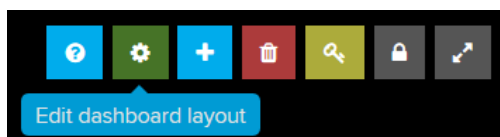
Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value
45

Published Value
0 bytes

Jika sudah terisi klik **Create block**.

Tahap terakhir, silahkan Anda atur posisi dari masing masing block dengan klik Edit dashboard layout, jika sudah sesuai kemudian Anda save



Nah untuk pembuatan antarmuka pada Adafruit sudah selesai.

PERHATIKAN !

Ganti/sesuaikan variabel berikut :

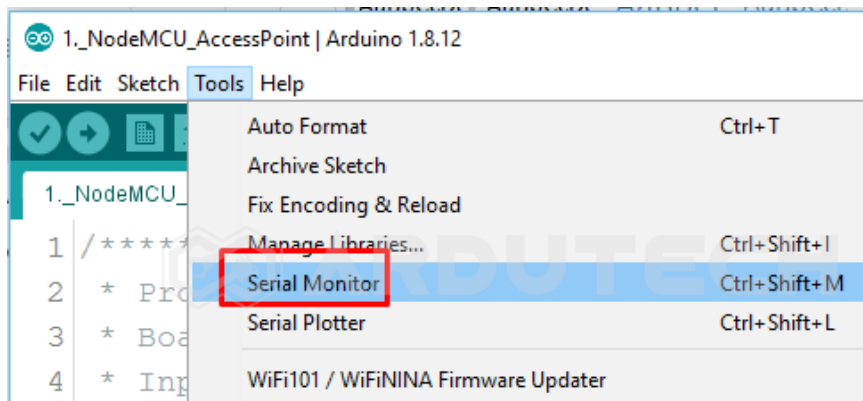
- Nama jaringan WiFi/hotspot : **WLAN_SSID**
- Password jaringan WiFi/hotspot : **WLAN_PASS**

- Username Adafruit IO : **AIO_USERNAME**
- Active Key Adafruit IO : **AIO_KEY**

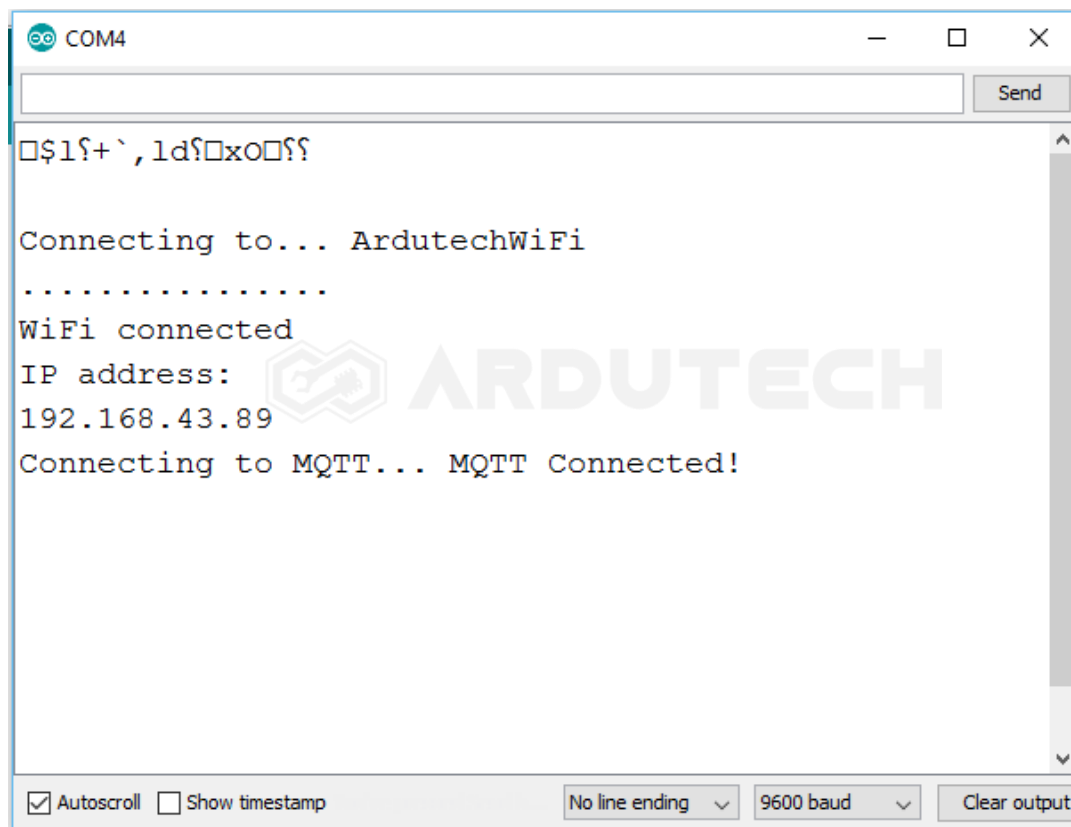
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

Jalannya Alat

Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools** → **Serial Monitor** kemudian seting baudrate pada nilai 9600.

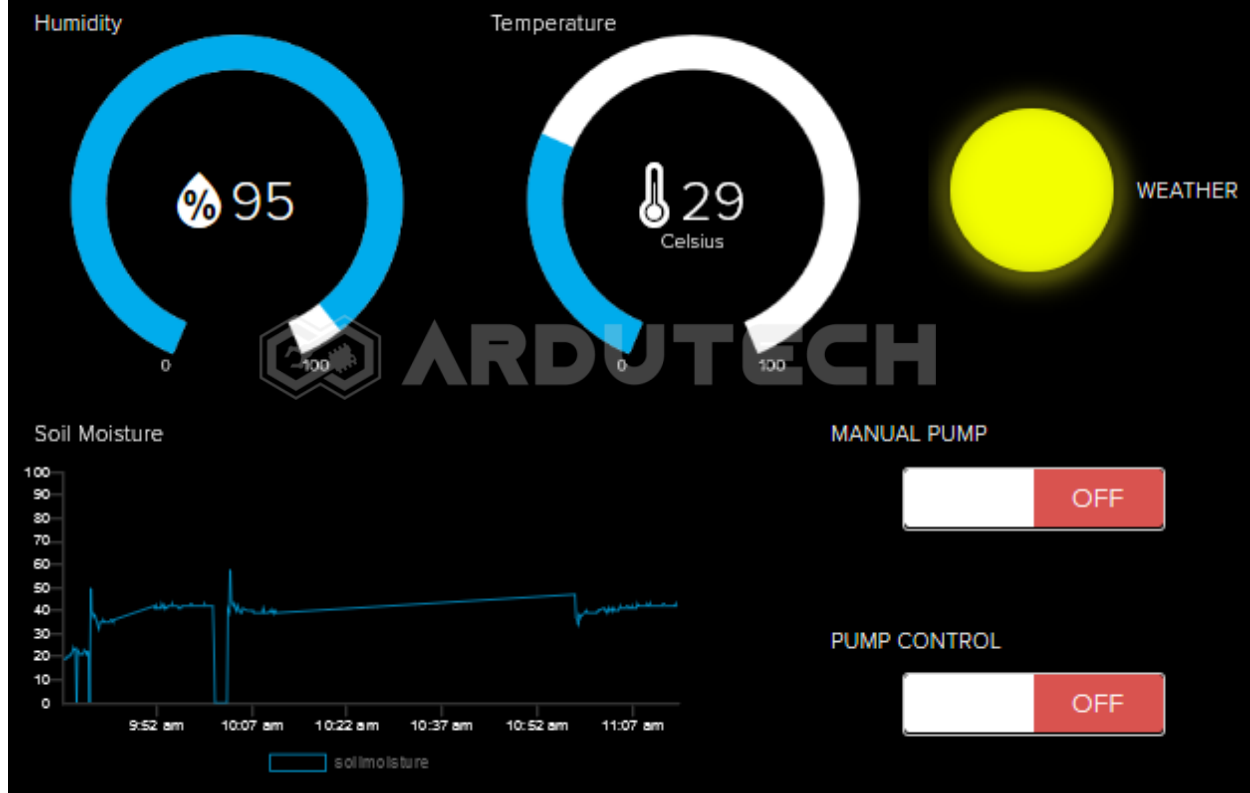


Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :

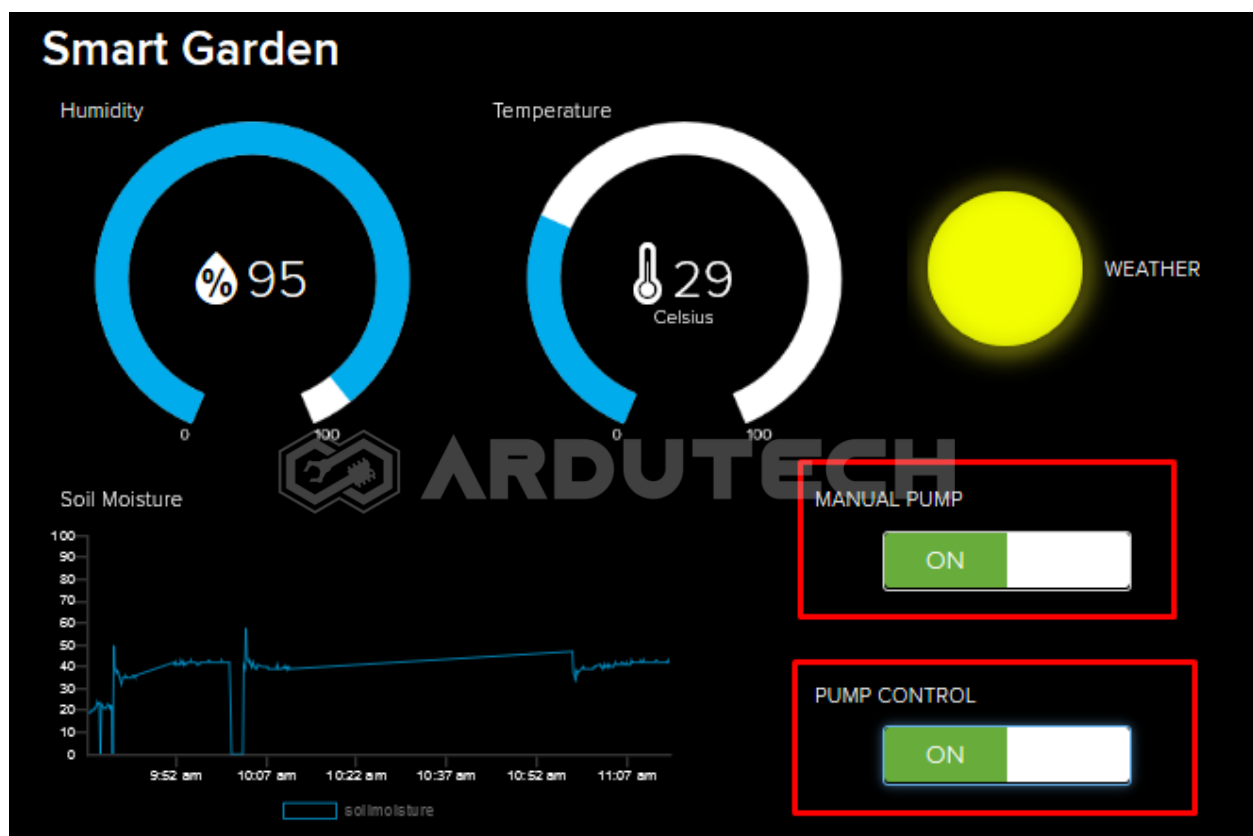


Setelah berhasil terhubung dengan jaringan WiFi sekarang kita kembali ke tampilan di Adafruit. Perhatikan nilai pada masing-masing blok menunjukkan hasil pembacaan masing-masing sensor.

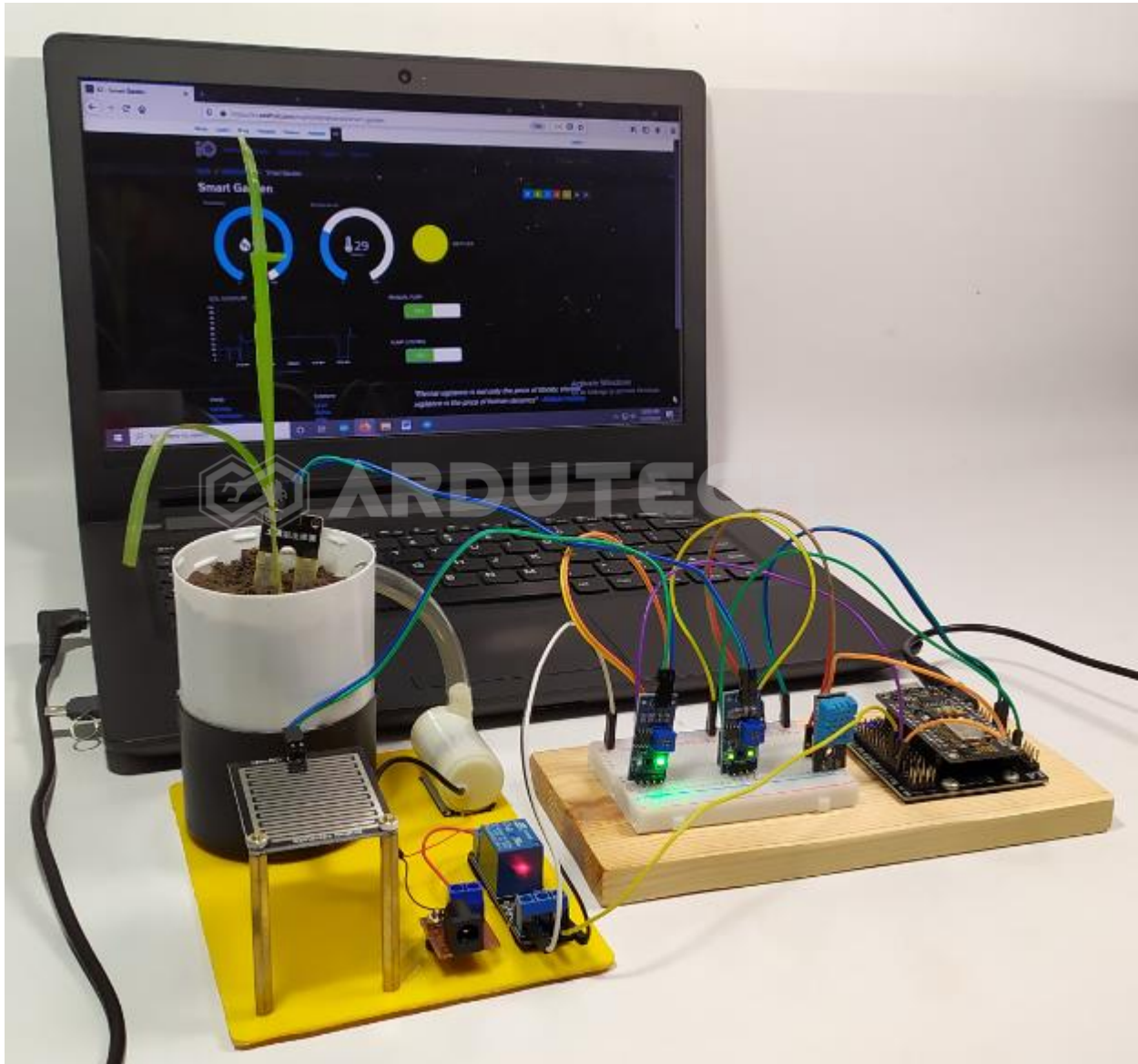
Smart Garden



Pada mode otomatis, beri perubahan kelembaban tanah, jika tanah kering (nilainya < 25) maka pompa otomatis akan ON dan akan mati kembali saat kelembaban tanah lebih dari 25. Kelembaban udara ditampilkan pada block gauge Humidity sedangkan suhu ruangan ditampilkan oleh block Temperature. Indikator weather menampilkan cuaca kebun dengan membaca sensor hujan, jika menyala kuning maka langit tidak hujan, sedangkan jika menyala biru tua maka terjadi hujan.



Pada mode manual pompa air dapat dikontrol ON/OFF-nya menggunakan switch Pump Control, sedangkan untuk pemilihan mode manualnya cukup posisikan switch Manual Pump ke posisi ON.



Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – “Sahabat Inovasi Anda”